SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour
QUESTIONS: 5 Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
   Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
   Estimate the probability of rolling a 6 using MLE.
5. Given:
   Input (x) = 4
   Weight (w) = 2
   Actual Output = 10
   Prediction:
   $y = w \times x$
   1. Find the predicted output.
   2. Calculate the error.
   3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

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1. Coin Toss – Maximum Likelihood Estimation (MLE)
Given:
Outcomes:
H H T H H H T H H T
Total tosses (n) = 10
Number of Heads = 7
Formula

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

Calculation

$$\hat{p} = \frac{7}{10} = 0.7$$

Answer
Estimated probability of getting Heads (p) = 0.7 (70%)

2. Students Passed Exam
Given:
Total students = 50
Students passed = 42
Formula

$$\hat{p} = \frac{\text{Passed}}{\text{Total Students}}$$

Calculation

$$\hat{p} = \frac{42}{50} = 0.84$$

Answer
MLE of passing probability = 0.84 (84%)

3. Defective Light Bulbs
Given:
Total bulbs = 100
Defective bulbs = 8
Formula

$$\hat{p} = \frac{\text{Defective}}{\text{Total}}$$

Calculation

$$\hat{p} = \frac{8}{100} = 0.08$$

Answer
MLE of defective probability = 0.08 (8%)

4. Biased Die
Given:
Total rolls = 60
Number of times 6 appeared = 18
Formula

$$\hat{p} = \frac{\text{Number of 6s}}{\text{Total Rolls}}$$

Calculation

$$\hat{p} = \frac{18}{60} = 0.30$$

Answer
Estimated probability of rolling a 6 = 0.30 (30%)

5. Weight Update Using Gradient Descent
Given

- Input (x) = 4
- Weight (w) = 2
- Actual Output = 10
- Learning rate (η) = 0.01
- Gradient = 4

1. Predicted Output
Formula:

$$\hat{y} = w \times x$$

Calculation:

$$\hat{y} = 2 \times 4 = 8$$

Answer: Predicted Output = 8

2. Calculate the Error
Formula:

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Calculation:

$$10 - 8 = 2$$

Answer: Error = 2

3. Update the Weight
Formula:

$$w_{\text{new}} = w - \eta \times \text{Gradient}$$

Calculation:

$$w_{\text{new}} = 2 - (0.01 \times 4)$$
$$w_{\text{new}} = 2 - 0.04 = 1.96$$

Answer

- **Predicted Output = 8**
- **Error = 2**
- **Updated Weight = 1.96**

Code of Conduct:

I M . Venkata Sai (192425223) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Venkata Sai

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation